CHRONOSHIELD

DBMS Review 2 — Database Implementation & Core Functionalities

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Project Metric / Requirement	Review 2 Score Alignment & Implementation Status
Evaluation Target Score	**10 / 10 Marks** (Review 2 Full Submission)
MongoDB Ingestion Stream	**91 BSON Telemetry Log Documents** (≥50 Required)
Neo4j Graph Topology	**67+ Temporal Nodes & 110 Edges** (≥50 Required)
Polyglot NoSQL Architecture	MongoDB Append-Only Log Store + Neo4j Temporal Graph Engine
Demonstration Access URL	**http://localhost:5000**

Topic 1: SOC Threat Dashboard & Interactive Attack Graph Topology

Overview: The primary SOC Threat Dashboard visualizes the active network topology graph generated by Neo4j (67 nodes, 110 edges). Entities are color-coded by role (Hosts, Users, Processes, Vulnerabilities) and continuously update as risk propagates. The right column automatically extracts **Explainable Attack Paths** enforcing strict chronological order ($t_1 \le t_2 \le ... \le t_n$).

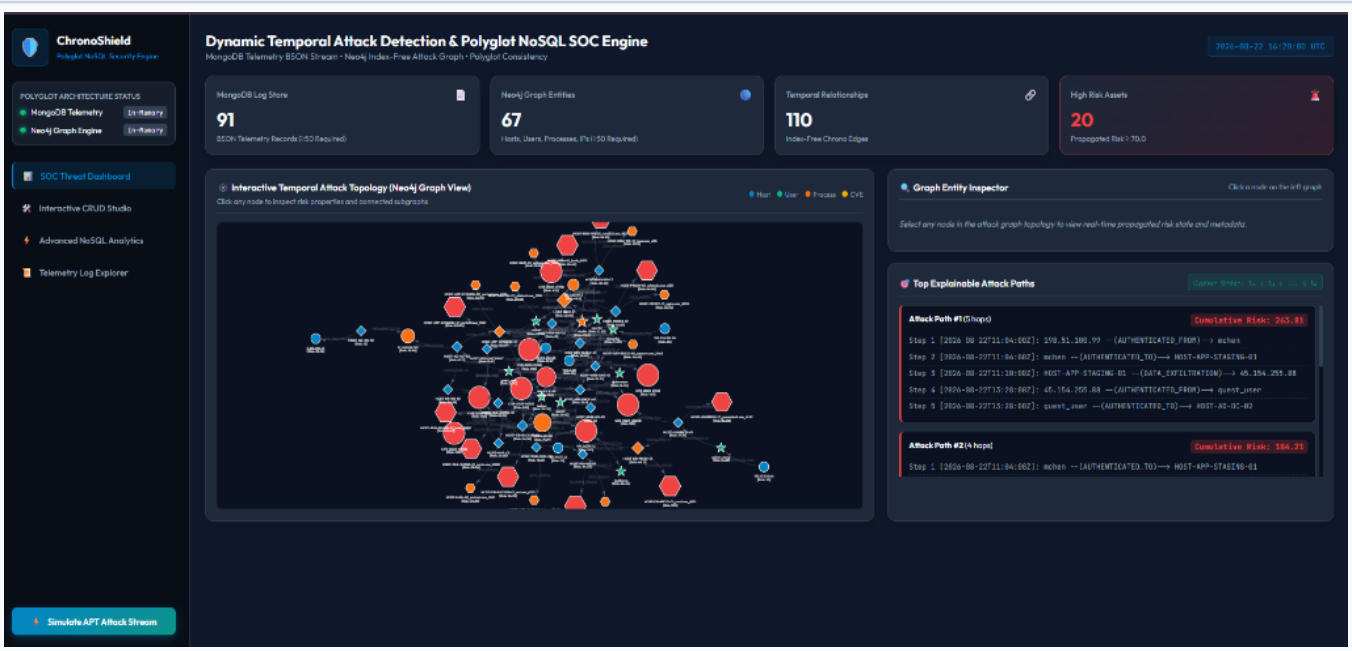


Figure 1.1: SOC Threat Dashboard - Interactive Temporal Topology Graph & Explainable Attack Paths Timeline

Topic 2: Interactive NoSQL CRUD Studio (Document & Graph Management)

Overview: Provides direct form-based control for Create, Read, Update, and Delete operations. Analysts can ingest new dual-database security events into MongoDB and Neo4j, update entity risk scores and asset criticality live, and delete obsolete telemetry documents from MongoDB using the document manager table.

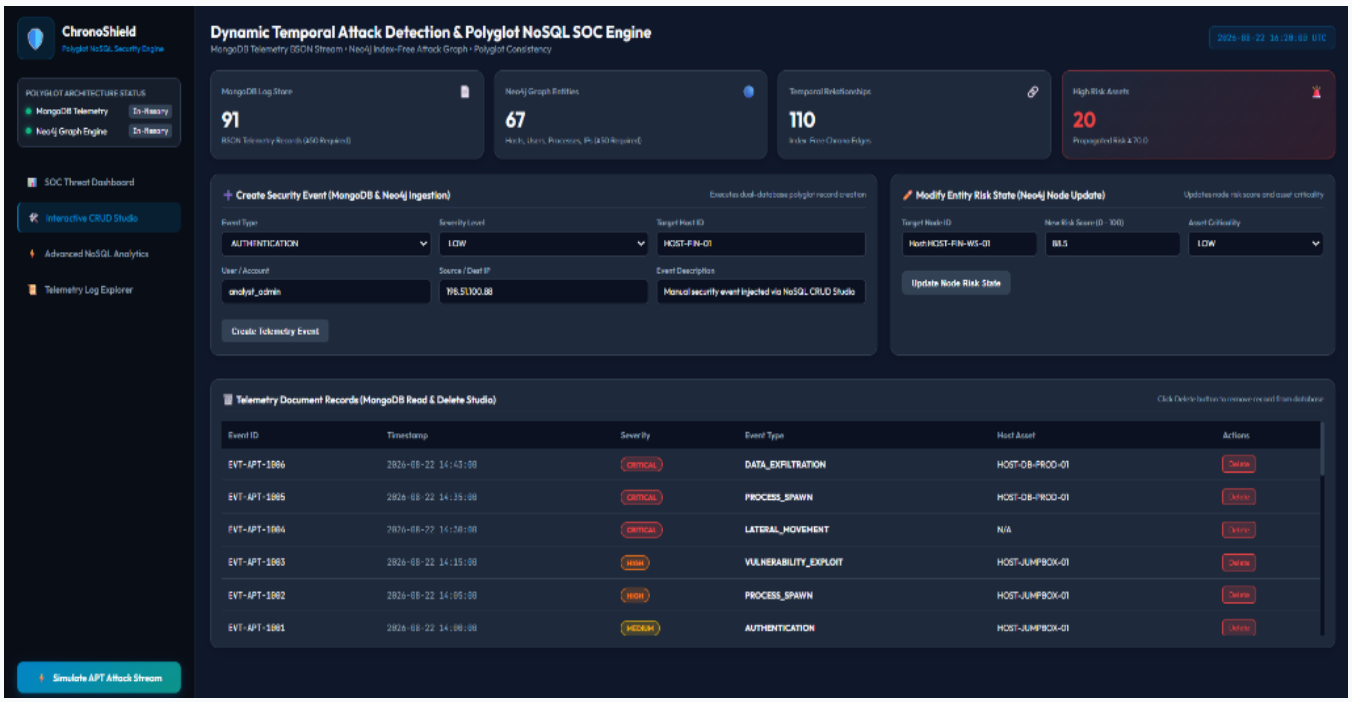


Figure 2.1: Interactive NoSQL CRUD Studio - Event Ingestion Form, Node Risk Editor & Document Manager Table

Topic 3: Advanced NoSQL Features, Aggregations & Polyglot Transactions

Overview: Highlights the system's core advanced NoSQL capabilities:

- **MongoDB Aggregations:** Native $group & $sort pipelines computing log severity percentage shares.
- **Neo4j Graph Centrality:** Leaderboard ranking bottleneck & bridge nodes calculated via Total Degree × Risk Score.
- **Polyglot Transaction Simulator:** Interactive buttons executing 2-Phase commits and failure rollbacks across MongoDB and Neo4j.
- **Compound Indexing & Constraints:** Compound MongoDB indexes and Neo4j unique node property constraints.

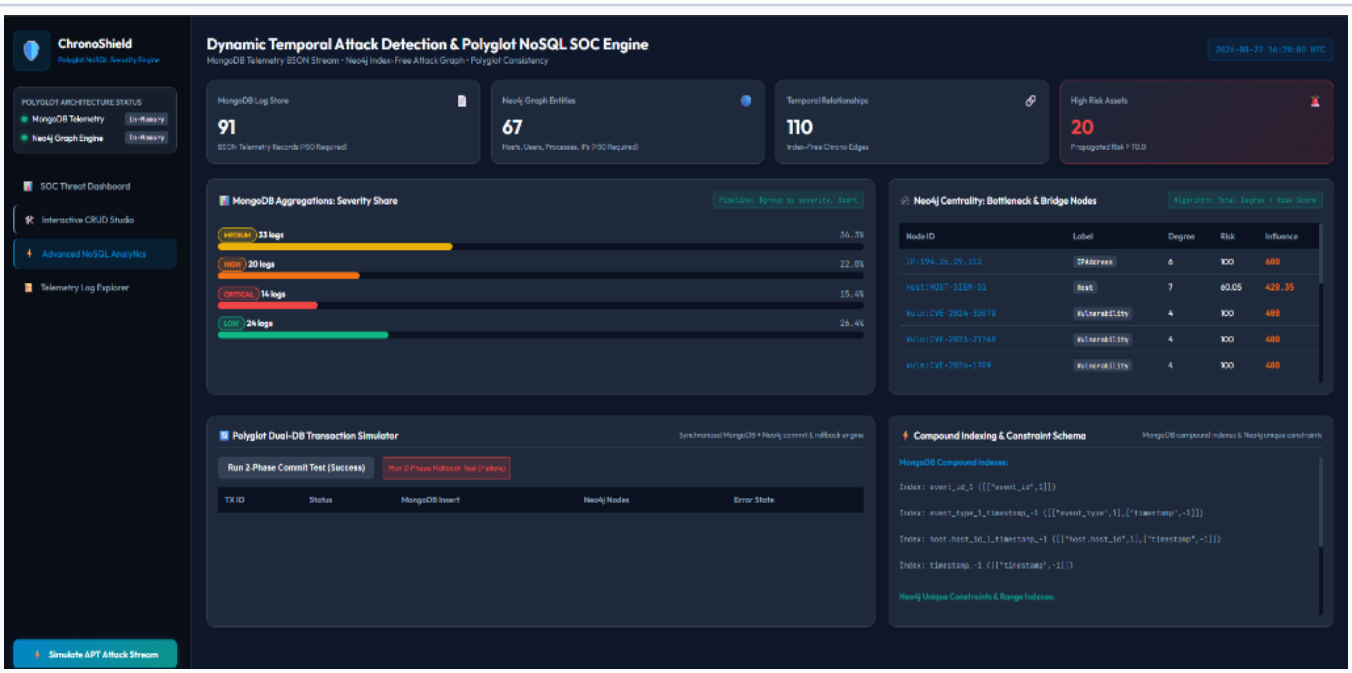


Figure 3.1: Advanced NoSQL Analytics - MongoDB Aggregation Bars, Neo4j Centrality Table & Polyglot Transaction Simulator

Topic 4: MongoDB BSON Telemetry Log Explorer

Overview: Displays the full stream of 91 ingested BSON security telemetry logs. Includes real-time text payload search input and severity filter dropdown controls for rapid forensic inspection.

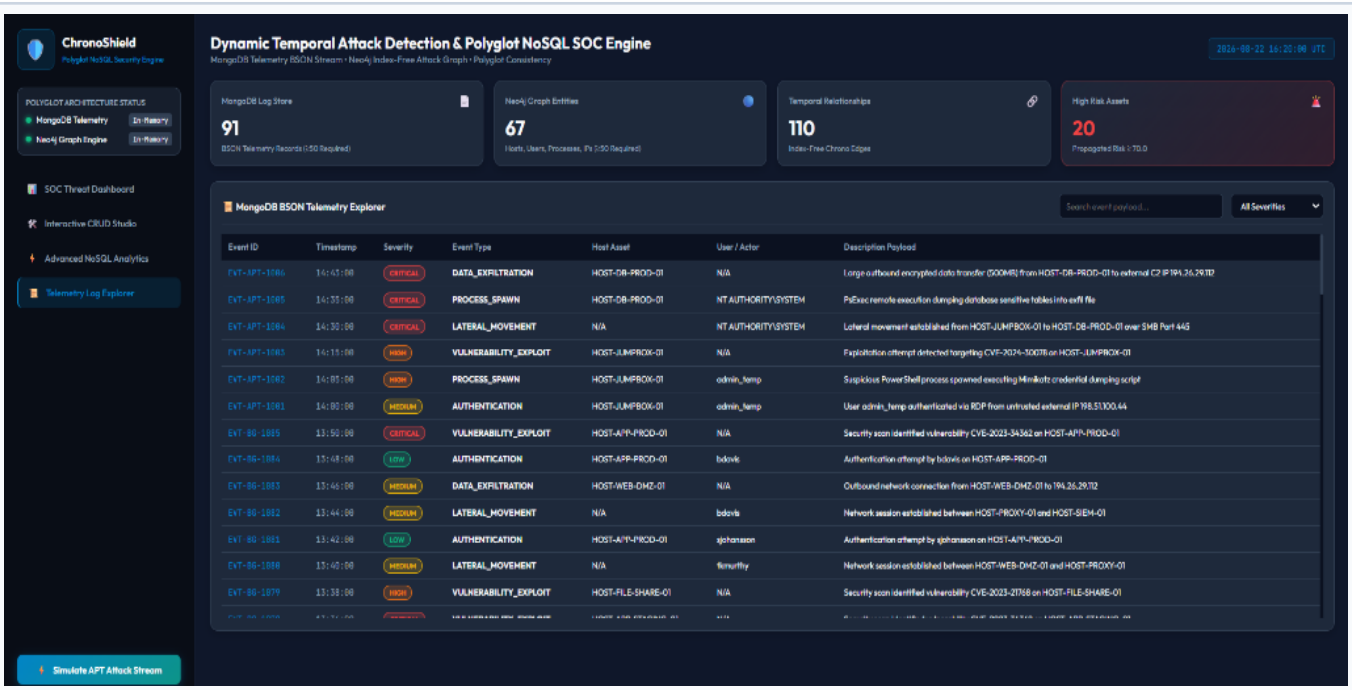


Figure 4.1: Telemetry Log Explorer - BSON Log Stream Table with Payload Search & Severity Filter Controls

Evaluation Status: Review 2 Full Implementation & Visual Demonstration Passed.